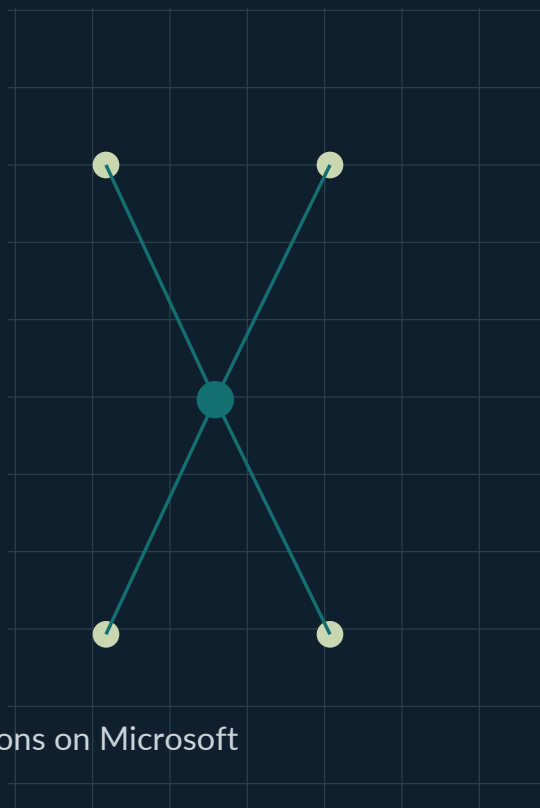


GRIDRESOLVE AI

Evidence before answers.

A governed multi-agent system for utility billing investigations on Microsoft Foundry.



MICROSOFT FOUNDRY / PROJECT ACCESS

WESTUS3

PROJECT ENDPOINT	<code>https://<foundry-resource>.services.ai.azure.com/api/projects/<foundry-project></code>
AZURE OPENAI	<code>https://<foundry-resource>.openai.azure.com/openai/v1</code>
PROJECT ID	<code>/subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.CognitiveServices/accounts/<foundry-resource>/projects/<foundry-project></code>

09

SPECIALIST AGENTS

14 / 14

PRE-SET CRITERIA

03

HOSTED RUNS

SUBMISSION OVERVIEW

One real synthetic high-bill dispute, a verified end-to-end workflow, an independently approved customer message, and a separate handoff for a human decision.

02 / THE CHALLENGE

A high bill is not one question.

The investigation spans numbers, equipment, policy, customer language, and financial authority.

CUSTOMER CONCERN / PARAPHRASED

"My bill is much higher. The meter must be broken. Can you confirm that and reverse the charge?"

FIVE DECISIONS HIDDEN IN ONE REQUEST

01

Billing

Do the charges and billed consumption reconcile?

02

Usage

Is there a plausible consumption change?

03

Meter

Do readings or diagnostics support a fault?

04

Policy

What can be explained or actioned under the rules?

05

Authority

What must a supervisor, not an agent, decide?

DESIGN POSITION

No unsupported equipment diagnosis. No promised financial adjustment. No self-approval by the agent that drafted the answer.

03 / SYSTEM ARCHITECTURE

Nine agents, separate responsibilities.

Foundry workflow v10 uses a shared conversation, explicit tasks, and deterministic release conditions.

INVESTIGATION / FOUR AGENTS

- 01 **CaseTriageAgent**
Classify the concern and scope the investigation.
- 02 **AccountEvidenceAgent**
Reconcile bills, register reads, and account records.
- 03 **UsageAnomalyAgent**
Separate observed patterns from unproven causes.
- 04 **PolicyKnowledgeAgent**
Retrieve the applicable synthetic policy constraints.

SYNTHESIS / TWO AGENTS

- 05 **ResolutionPlannerAgent**
Build the plan and classify material claims.
- 06 **CustomerCommunicationAgent**
Draft the customer-facing explanation.

CONTROL / THREE AGENTS

- 07 **EvidenceComplianceAgent**
Independently approve or withhold release, with reasons.
- 08 **EscalationCoordinatorAgent**
Prepare a decision package when review is required.
- 09 **CaseAuditAgent**
Record disposition and provenance for both routes.

Separation of duties: the communication agent writes the response; the compliance agent decides whether it can be released. Human follow-up is a separate decision.

04 / DESIGN MECHANICS

Claims need a source, not confidence alone.

The architecture makes facts, assumptions, policy constraints, and decisions inspectable.

THE CONTROLLED CASE RECORD

EVIDENCE

What was observed

Bills, dates, reads, diagnostics, source references.

POLICY

What action is allowed

Rule IDs, versions, review and disclosure requirements.

CLAIMS

What may be said

Supported status, linked evidence and policy IDs.

FROM REQUEST TO AUDITED OUTCOME



FOUNDRY IMPLEMENTATION

- **Input discipline.** Each node receives a literal task message. This corrected the earlier agent stalls.
- **Strict schemas.** Evidence, policy, communication and audit outputs are required to be structured.
- **Two gates.** Release approval is distinct from whether the underlying case needs human follow-up.

05 / WORKED EXAMPLE

The meter claim did not survive the evidence.

Synthetic case SYN-CASE-4003, executed on the hosted GridResolveAIWorkflow v10.

ACCOUNT QUESTION

\$152.80 to **\$203.40**

The customer wants the system to confirm meter failure and reverse the bill.

RECORDS RECONCILED

JUNE 30 READ

41,820 kWh

JULY 31 READ

42,690 kWh

JULY BILLED

870 kWh

42,690 - 41,820 = 870 kWh

Register change equals billed consumption for the period.

WHAT THE ACTUAL RELEASE SAID

"The meter register increased from 41,820 kWh on 2026-06-30 to 42,690 kWh on 2026-07-31, a change of 870 kWh. That register change matches the 870 kWh billed for July 2026."

"We did not find evidence in the supplied diagnostics or event logs that the meter over-recorded usage."

The important distinction: the message also notes the limits of remote diagnostics. It does not prove the meter can never fail, and it does not promise a credit.

06 / GOVERNANCE

Approval of a message is not closure of a case.

The final hosted run passed compliance and still sent the open decision to a person.

TWO INDEPENDENT DECISIONS

GATE A / COMMUNICATION

APPROVE

Compliance found the explanation supported, policy-grounded, and fit for release. The approval included reasons and zero failed checks.

GATE B / FOLLOW-UP

PENDING HUMAN REVIEW

Billing Supervisor receives the open inspection or adjustment decision. A safe customer message does not grant financial authority.

REVIEW PACKAGE / OBSERVED IN THE FINAL RUN

DESTINATION

Billing Supervisor

DECISION REQUIRED

Human review to determine whether to perform on-site inspection and/or authorize financial adjustment.

WHAT THIS DEMONSTRATION DOES AND DOES NOT ESTABLISH

- Observed on v10: nine completed agents; a reasoned approval; readable release; separate human handoff; audit aligned with platform records.
- Not observed on v10: the non-approval branch, an automated correction loop, or a case that requires no human follow-up.
- Run 2 withheld output and escalated, but its exact escalation reason was not established.

07 / RUNTIME HISTORY

The defects are part of the evidence.

All three hosted executions and their original evidence folders are retained for review.

THREE REAL FOUNDRY RUNS

OBSERVED	RUN 1 / v6	RUN 2 / v9	FINAL / v10
Agents completed	7 of 8	6 of 9	9 of 9
Evidence ledger	None	None	22 entries
Customer output	Raw expression	Withheld	Readable
Compliance	Approved	Escalated*	Approved
Human follow-up	Not built	Package	After release
Audit discrepancies	7	6	0
Est. run cost	\$0.0334	\$0.0292	\$0.0685

* Run 2 escalated with no stated reason. It must not be presented as proof that the system identified a specific unsupported claim.

WHAT CHANGED THE OUTCOME

The platform record exposed missing task messages on the shared conversation. Version 10 added explicit invocation, strict outputs, and separate routing gates. Its 14 acceptance criteria were fixed before the third run.

08 / VERIFICATION

A pass has to name the test it passed.

Runtime acceptance, local checks, platform telemetry, and evaluation are reported separately.

14 / 14

pre-registered runtime criteria

1,174

passing local checks

94 / 94

read-only live configuration

LOCAL VERIFICATION BREAKDOWN



OTHER CHECKS AND LIMITATIONS

- 105 / 105 cases executed on the actual Power Fx engine.
- 11 / 11 Agent Framework replay parity checks, local only.
- 12 / 13 deterministic checks passed on the final run. The prepared root-cause label did not match the planner label; the check remains failed.

09 / OPERATIONS AND COST

Measure the platform, not only the narrative.

The runner was cross-checked against Azure Monitor and the available trace records.

EVIDENCE FROM THE THREE REAL EXECUTIONS

26**model requests**

Across all hosted runs

61**dependency spans**

Platform tracing

186,614**input tokens**

Matches Azure Monitor

42,155**output tokens**

Matches Azure Monitor

COST REPORTED WITHOUT GUESSING THE BILL

PROVISIONAL, CALCULATED FROM MODEL USAGE

\$0.1311

Three runs at published list rates. Azure Cost Management had not yet shown the billed amount. This is an estimate, not a confirmed invoice.

PRODUCTION SECURITY STATUS

- OBSERVED** Synthetic-only demonstration; no external agent tools or real utility APIs.
- CONFIGURED** Microsoft.DefaultV2 guardrail attached; 26 checks observed, zero blocked.
- OPEN** Public network access and key authentication remain enabled; message content recording is on.

10 / PRODUCTION PATH

A working prototype, not a utility deployment.

The next build steps are intentionally separated from what the hosted demonstration proved.

ALREADY IMPLEMENTED LOCALLY

- **Typed integration boundary**

Eight ports and synthetic adapters; 138 local tests.

- **Approval separation**

An agent principal cannot authorize a financial adjustment.

- **Migration proof**

The v10 workflow replayed in Microsoft Agent Framework with 11 of 11 parity checks.

REQUIRED BEFORE ANY REAL CUSTOMER USE

01

Connect governed sources

Approved billing, meter, policy and case APIs; identity, freshness and access controls.

02

Move orchestration

Hosted Foundry Workflows retirement is announced for 2026-12-01. Deploy the Agent Framework path with durable state.

03

Exercise unproven branches

Run other cases, reject and correction routes, and no-follow-up outcomes with permission and budget.

04

Harden operations

Restrict network and key access, govern recorded content, add alerts, budgets and human work-queue delivery.

11 / EVIDENCE AND STATUS

Judge the system by the record.

An architect submission with one hosted end-to-end success and a clear account of what remains.

WHAT THIS PROJECT DEMONSTRATES

A multi-agent workflow can reconcile a contested bill, refuse to present an unsupported meter-failure allegation as fact, release a grounded explanation, and retain human authority over the next decision.

SUPPORTING ARTIFACTS

01 Hosted run and acceptance plan

evidence/runtime/20260921T000142Z_SYN-CASE-4003_c2be2b51/ | docs/FINAL_RUN_ACCEPTANCE_PLAN.md

02 Final run results and current status

docs/FINAL_RUN_RESULT_2026-09-20.md | docs/CURRENT_STATUS.md

03 Live platform telemetry

evidence/platform_telemetry/ | 26 model requests, 61 dependency spans

04 Evaluation and replay

evaluation/ | migration/agent_framework/ | tests/powerfx_gate

05 Foundry and Control Center captures

evidence/screenshots/ | evidence/app-screenshots/ (offline UI is labeled as such)

SCOPE STATEMENT

Synthetic data only. One successful end-to-end hosted run on v10. No real utility integrations, no confirmed billed amount, and no claim that untested branches or repeatability are proven.